MohammadAmin Sadeghi

Architectural Designer | Data Center Design, BIM & AI-Augmented Workflows

Tehran, Iran | +98 993 434 8489 | sadeghimohammadamin83@gmail.com | linkedin.com/in/aminetun | GitHub | Portfolio

Professional Summary

Architecture undergraduate and architectural designer specializing in **data center planning**, **BIM**, and **AI-augmented design workflows**. Has laid out **120+ rack positions** across six data center projects, from a 20 ft modular container to a 48-rack technical level with eight 20-ton cooling units, mostly as **retrofit** inside existing halls with fixed grids and slab levels. Uses **AI agents as a working team**: writes precise briefs, splits work into reviewable steps, and verifies every output against drawings, sources, and real dimensions. Has directed **12+ AI-assisted projects** this way, spanning GIS research, site-analysis decks, code-generated 3D models, a data center monitoring platform, a BIM object library, a desktop app, and a cross-platform game. The result is the output of a small studio, delivered by one designer, without losing technical accuracy.

AI-Augmented Design & Development

- **Method:** Treats AI agents (**Claude Code**, **ChatGPT**) as a delegated team. Writes briefs with constraints, real dimensions, and acceptance criteria, breaks large jobs into small checkable steps, and reviews each result before moving on. Nothing is issued until it is checked against the drawing or the source.
- **Accuracy rules for AI output:** Requires every figure to carry its source, removes any number that cannot be traced, and records data gaps openly instead of letting the model fill them, applying the same observed / documented / derived discipline used in data center proposals.
- **Agent operations:** Runs parallel agent sessions in the cloud, on desktop, and from mobile, keeps all work versioned in **GitHub** repositories with branches and releases, and extends agents with **MCP connectors** (Figma, GitHub) and installed agent skills.
- **Research & GIS:** Directed the Ramsar urban baseline study end to end with AI: an Excel database built from **40 sources**, GIS mapping scripts and 15 maps, a 77-page report, and 38 slides, with **5 data gaps** documented rather than guessed.
- **Presentations:** Produced site-analysis decks for a Physical Education faculty in Tehran (with an interactive 3D site model) and for Varish village (a 50-page PDF with a clickable table of contents). Translated two decks (14 and 23 slides) into Persian with right-to-left layout, and rebuilt a university architecture presentation with 18 new images after removing every unsourced figure.
- **3D & CAD by code:** Generated an interactive 3D viewer with SVG/PNG drawings for a villa terrace-and-stair reconstruction, a **parametric OpenSCAD** watch case (STL plus fit-test ring) for 3D printing, an interactive **Three.js** model of a sports complex, and DXF-to-A3 sheet workflows.
- **Software from a design brief:** Took **ECHORA** from concept to version 0.5.4 (Windows and Android builds, bilingual website, release notes) and built **MINDSPACE**, an Electron desktop app with semantic search and source-cited AI answers. Directed architecture, visual identity, testing, and releases through AI agents, and prototyped **Amin DCI** and **KautBIM** with AI app builders.

Experience

Architectural, Data Center & Digital Designer | *Padisar Informatics* *May 2026 – Present*

- Designed architectural and technical layouts for **data center** and **telecom infrastructure** projects, including rack plates, **hot/cold-aisle containment** strategies, and service-route coordination inside existing buildings.
- Delivered the **Two-Level Telecom Facility** (issued for construction, three-person team) and the **48-rack Technical Level** proposal, both shown under Data Center Projects.
- Developed **BIM models** and coordinated drawing packages in Revit, covering plans, sections, schedules, and room data, plus 3D visualizations and **animated assembly sequences** in Enscape and 3ds Max.
- Wrote client-facing design proposals and technical documentation that mark every value as **observed, documented, or derived**, keeping established data separate from open items.
- Designed and built corporate web pages in **WordPress**.

Freelance Designer | *Selected projects*

- Data hall layout studies, BIM documentation, visualization, and design proposals for clear technical briefs.

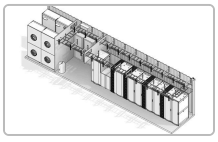
Instructor, Adobe Design Software | *Design software courses*

- Taught **Photoshop, Illustrator, InDesign, and Adobe XD** with practical exercises, project briefs, and critiques.

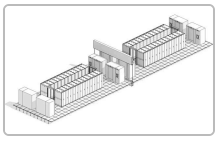
Technical and Administrative Specialist | *Sanat Pajouhan Matrah*

- Technical reports, inspection documentation, and structured records in Excel and Word.

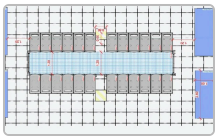
Data Center Projects



Modular Data Center · *20 ft container · Revit, Enscape, AutoCAD, 3ds Max · 2026*
Modeled a complete containerized facility at true size: cabinet row, in-row cooling, UPS, electrical panel, cable ladder, CCTV, and external condensers, so clashes surfaced before fabrication. Each system was exported as a layer for exploded axonometrics and an assembly sequence.



Technical Level, 48 Racks · *Padisar Informatics · 19-page proposal · Aug 2026*
One 16.60 × 29.67 m level holding two contained technical rooms, 48 racks, and eight 20-ton cooling units. A four-step axonometric sequence made it readable for non-technical approvers while staying usable for the engineers.



Two-Level Telecom Facility · *Padisar Informatics · issued for construction · Aug 2026*
Part of a three-person team. Designed a 12-cabinet central pod with a 1.20 m contained cold aisle, four in-room cooling bays, and a 10-position perimeter row on a five-line column grid.

Existing Hall Conversion | *Revit* *2026*

- Converted a single-storey hall (27.33 m on the lower grid line, ≈131 m²) into a data center space with one **contained pod of 24 rack positions** and a 1.20 m aisle. Coordinated clearances and space for HVAC and electrical against a fixed column grid.

Rack Room Reorganization | *Revit*, *AutoCAD* *2024*

- Reorganized an in-use room into a structured **24-rack hall** (two rows of 12, cooling units at both flanks, 1.20 m service clearance). Documented it with same-viewpoint before/after axonometrics and an issued 1:50 reflected ceiling plan (A1.01, May 2024), with design credit on the issued drawing.

Data Hall Retrofit | *Revit*, *AutoCAD*, *IFC*, *Enscape* *2026*

- Fitted **24 cabinets** in two facing rows with a 1.20 m contained aisle and four plant units into an existing 24.34 m floor plate. A glazed perimeter technical corridor gives access to every cabinet face without entering the contained aisle.

Digital Products

Amin DCI | *Data center monitoring and operations platform* *Web*

- Designed and built a platform that inventories sites, rooms, racks, and devices, shows **floor plans with physical rack layout**, live alerts by severity, power (IT, facility, **PUE**) and environmental data, and maintenance workflows. Integrates with **NetBox** and **Zabbix** and computes a 0–100 health score.

KautBIM | *BIM and 3D object library* *Web*

- Built a library of dimensionally accurate BIM objects with structured parameters, organized by discipline (data center, HVAC, electrical, mechanical). Formats include RFA, RVT, FBX, OBJ, and SKP for Revit, SketchUp, 3ds Max, Blender, AutoCAD, and Rhino.

Architecture & Urban Planning Projects

The Last Shadow of Time (Museum) | *Revit*, *Enscape*, *3ds Max*, *Photoshop*, *InDesign* *2026*

- Designed a museum of memory, time, and light on a **radial axis around a central void** and carried it from concept and massing through plans, 1:200 sections, a rotated vertical-fin façade, full visualization, and two presentation boards.

Outdoor Tennis & Volleyball Complex | *3D modeling*, *Three.js*, *OBJ / GLB* *2026*

- Planned a 60 × 100 m sloping site (4 m fall) with both courts sunk into the slope. Spectators arrive from the high side and players from the low side, so the two circulation systems stay separate from gate to court. Delivered a 1:400 site plan, sections at 1:250 and 1:150, and an interactive 3D model.

Ramsar Urban Baseline Study | *GIS*, *Excel* *2026*

- Analyzed population, climate, terrain (−29 to 3,461 m within 36 km), hazards, land cover, movement, and tourism from official data and GIS layers, and synthesized them into six site principles. Delivered a **77-page report, 38 slides, and 15 GIS maps**.

Technical Skills

Data Center Design: Data hall layout, rack & containment planning, cooling coordination, raised-floor setting-out, clearance & circulation verification, as-built documentation
Architecture: Architectural planning, retrofit & conversion, site & existing-condition analysis, drawing-set production, technical documentation
BIM & CAD: Revit, AutoCAD, SketchUp, 3ds Max, IFC / DXF workflows
Visualization: Enscape, D5 Render, axonometric & cutaway diagrams, animated assembly sequences
Graphic & Digital: InDesign, Photoshop, Illustrator, Adobe XD, UI/UX design, WordPress, HTML & CSS
Research & Data: GIS mapping, Excel, urban baseline analysis
AI & Automation: Claude Code, ChatGPT, prompt & brief writing, MCP (Figma, GitHub), agent skills, Three.js, OpenSCAD, Git / GitHub, Electron, React

Education & Languages

B.Arch in Architecture | *Islamic Azad University, Central Tehran Branch (Poonak)* *In progress*
Persian: Native | **English:** Upper-intermediate